

AI-DRIVEN WEARABLE TECHNOLOGY FOR ROUTINE MONITORING IN ELDERLY WITH DEMENTIA

Mariana Carvalho (mccarvalho@ipca.pt) MIAA

Vítor Carvalho (vcarvalho@ipca.pt)

Pedro Morais (pmorais@ipca.pt)

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Abstract

The aging global population and rising life expectancy has led to a significant rise in chronic diseases and disabilities, with dementia being one of the most prevalent and challenging conditions. It affects memory and cognitive functions, disrupts daily activities, and increases dependency, placing greater strain on caregivers. Monitoring changes in daily routines is crucial for early intervention and ensuring appropriate care (Rocha et al., 2024).

This research proposes an intelligent system capable of human activity recognition, including mobility, sleep, hygiene, meals, and fall detection, as well as the identification of routine changes in elderly individuals with dementia. The system will rely on data collected from wrist-worn wearable sensors (Rocha et al., 2024), providing continuous and non-invasive behavioral monitoring. A key challenge is the variability across individuals and limited availability of real data from people with dementia (Carvalho et al., 2025). To address this, the methodology begins with data from collection from healthy volunteers underrepresented activities, such as hygiene routines. Various deep learning architectures are being evaluated for their suitability in handling time-series sensor data. Models will be assessed using standard classification metrics and their capacity to detect routine deviations, prioritizing low false alarm rates and minimizing missed anomalies.

At the current stage, data preprocessing is complete and baseline models for activity classification have been implemented. The next phase involves simulating routine deviations to validate the system's anomaly detection. The expected outcome is a robust and adaptable model that can be embedded in a wearable prototype, enabling more effective, autonomous, and personalized monitoring for elderly individuals living with dementia.

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